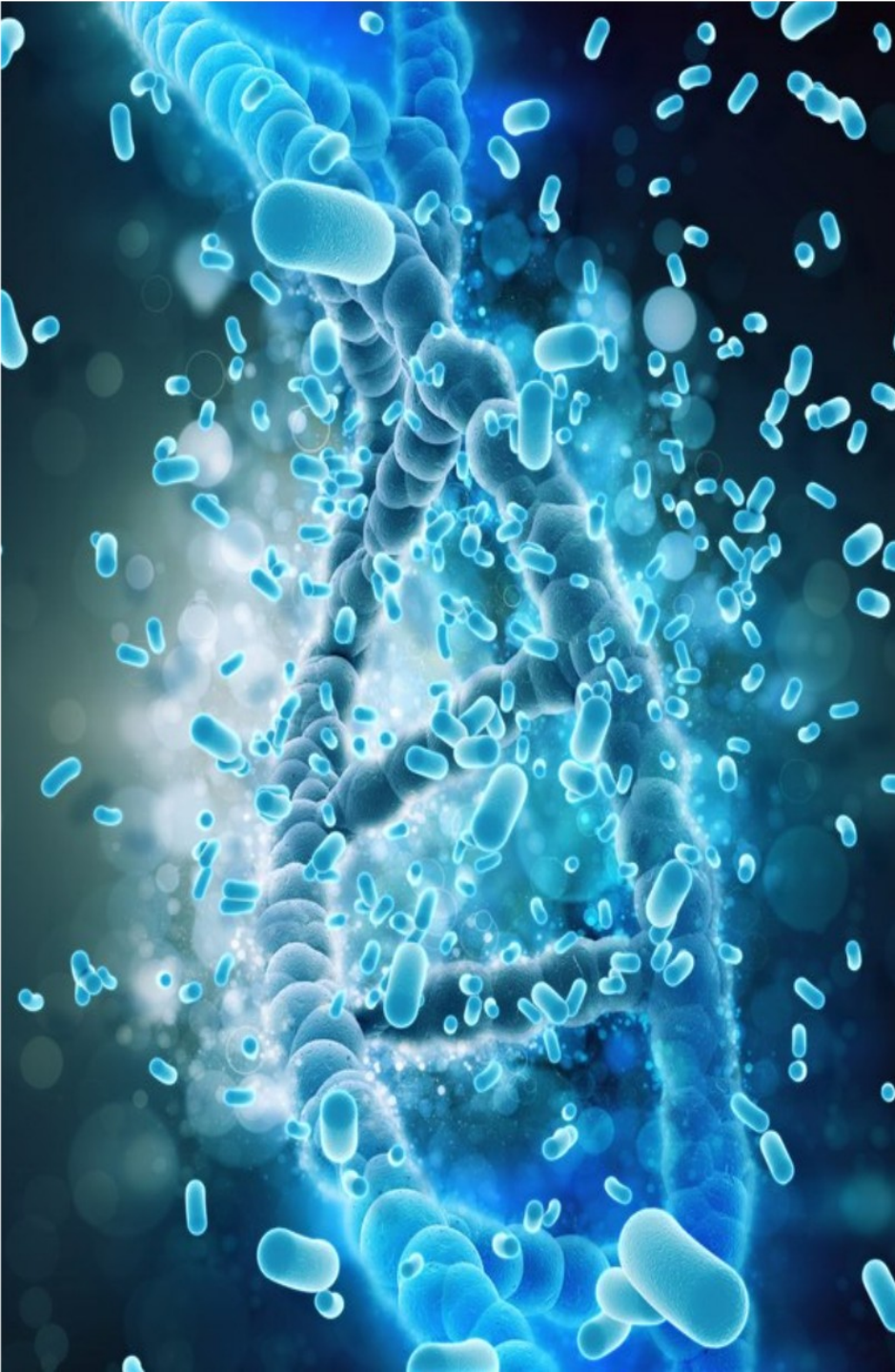




Microbial Genetics

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Microbial Genetics

Microbial genetics is a field of biology that studies the genetics of microorganisms:

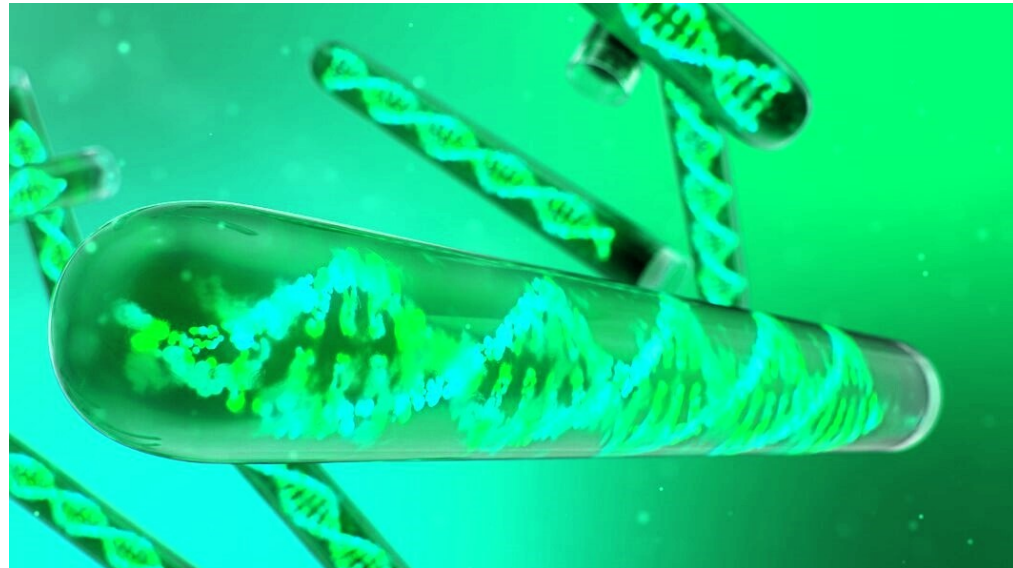
- Prokaryotes
- Archaea
- Viruses
- Fungi
- Protozoa.

Genetic Materials in Microorganisms

The structure and function of DNA and RNA as the carriers of genetic information is diverse among microorganisms

Genome organization:

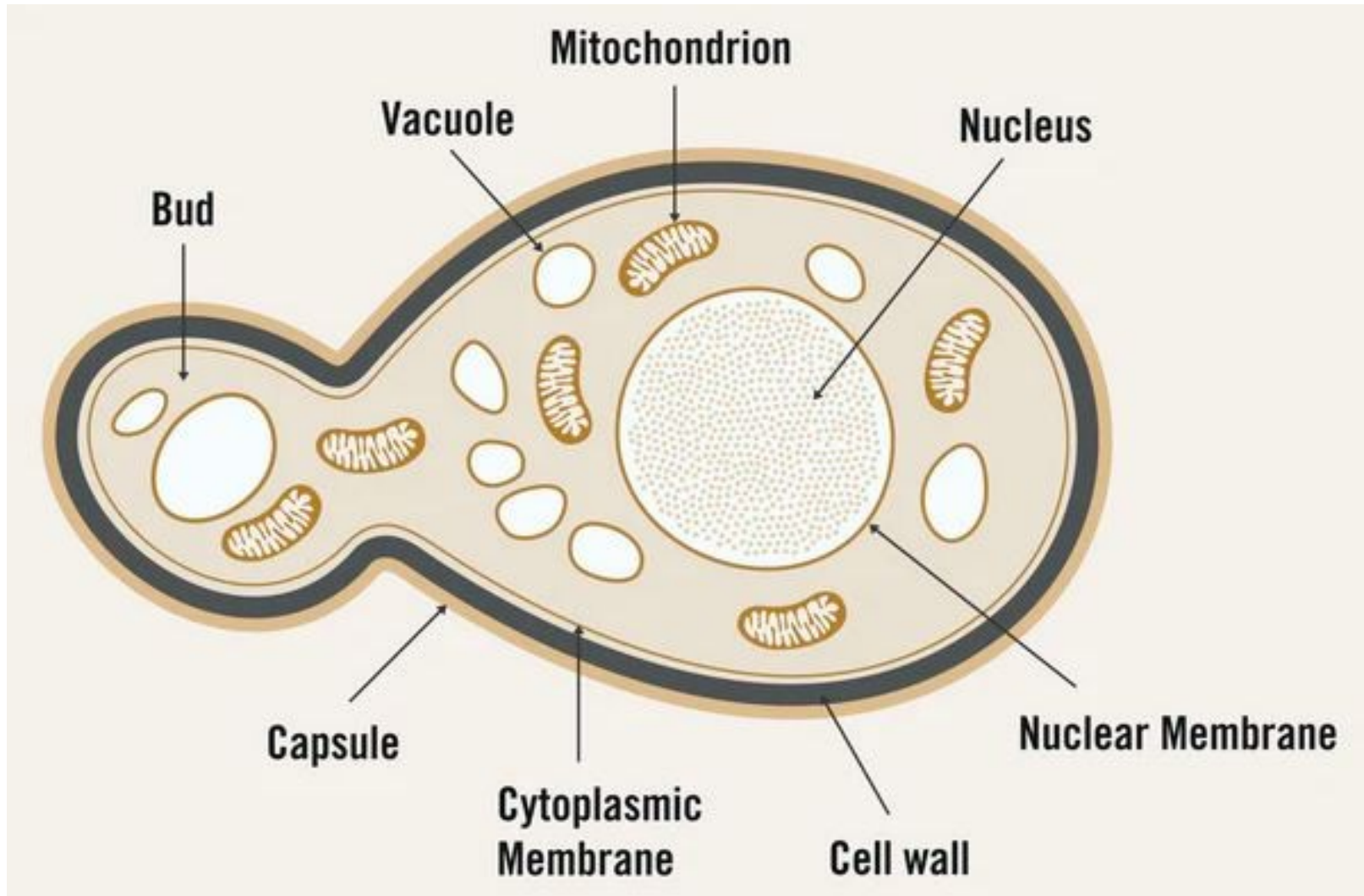
- Prokaryotic genomes (circular chromosomes, plasmids).
- Viral genomes (DNA or RNA, single-stranded or double-stranded).
- Eukaryotic microbial genomes (linear chromosomes, mitochondria, or chloroplast DNA).



Fungal Genome

- **Fungal genome locations:** (Nuclear and Mitochondrial genome)
 - The fungal chromosomes are located inside nucleus.
 - The number varies from $n=3$ to $n=20$ chromosomes
 - Few genes are located inside **mitochondrial** DNA
- **Size and structure:** Fungal genomes vary widely in size and complexity, typically ranging from 10–100 Mb.
- **Chromosome organization:** Fungi have linear chromosomes, often with well-defined centromeres and telomeres.
- **Genetic diversity:** Fungi exhibit high genetic variation due to their diverse reproductive and mutational mechanisms.
- **Repetitive DNA:** Many fungal genomes contain repetitive sequences, including transposable elements, which influence genome evolution and plasticity.

Nuclear and Mitochondria

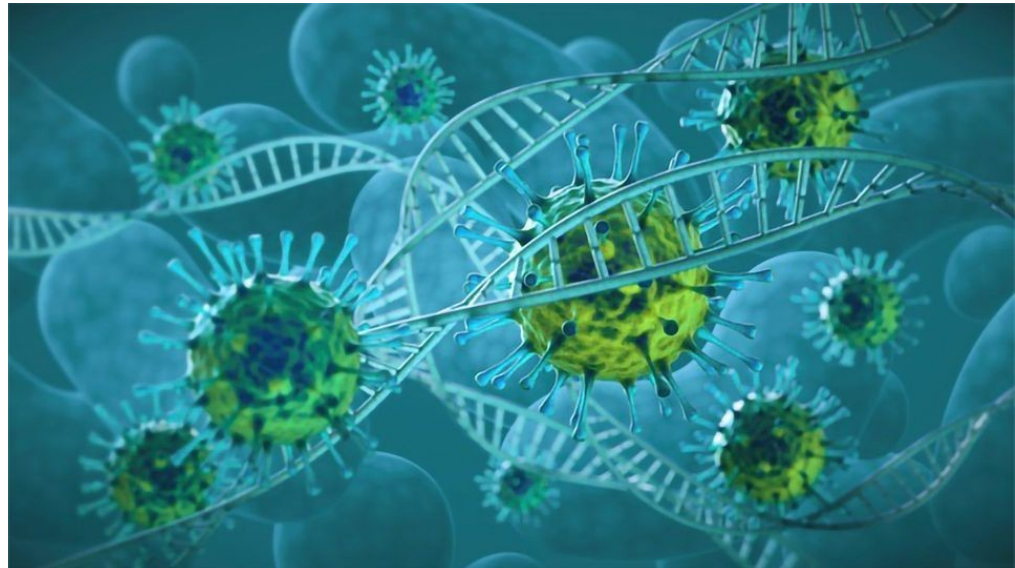


Viral Genetics

- **Viral genetics:** refers to the study of the genetic material of viruses, how they replicate, mutate, evolve, and interact with host cells.
- **Type of genome:** Virus genome is either DNA or RNA located in a protein cover (**capsid**) or combined with the cover (**Nucleocapsid**)
- **Segmented genome:** Viruses like influenza have genomes split into multiple segments, allowing reassortment and rapid evolution.
- **Genetic Diversity:**
 - Viruses exhibit significant genetic diversity and unique mechanisms of genetic exchange
 - Viruses evolve rapidly due to their high mutation rates, particularly RNA viruses (because RNA polymerases lack proofreading ability)

Applications of Viral Genetics

- **Vaccine design:** Understanding genetic changes helps predict emerging strains (e.g., annual influenza vaccines).
- **Antiviral therapies:** Targeting specific viral genes or proteins.
- **Gene therapy:** Using modified viruses to deliver therapeutic genes to cells.
- **Epidemiology:** Tracking viral mutations and outbreaks (e.g., genomic sequencing of SARS-CoV-2).



Prokaryotic Genome

The prokaryotic genome refers to the complete set of genetic material in a prokaryotic cell, including chromosomal and extrachromosomal DNA.

Structure:

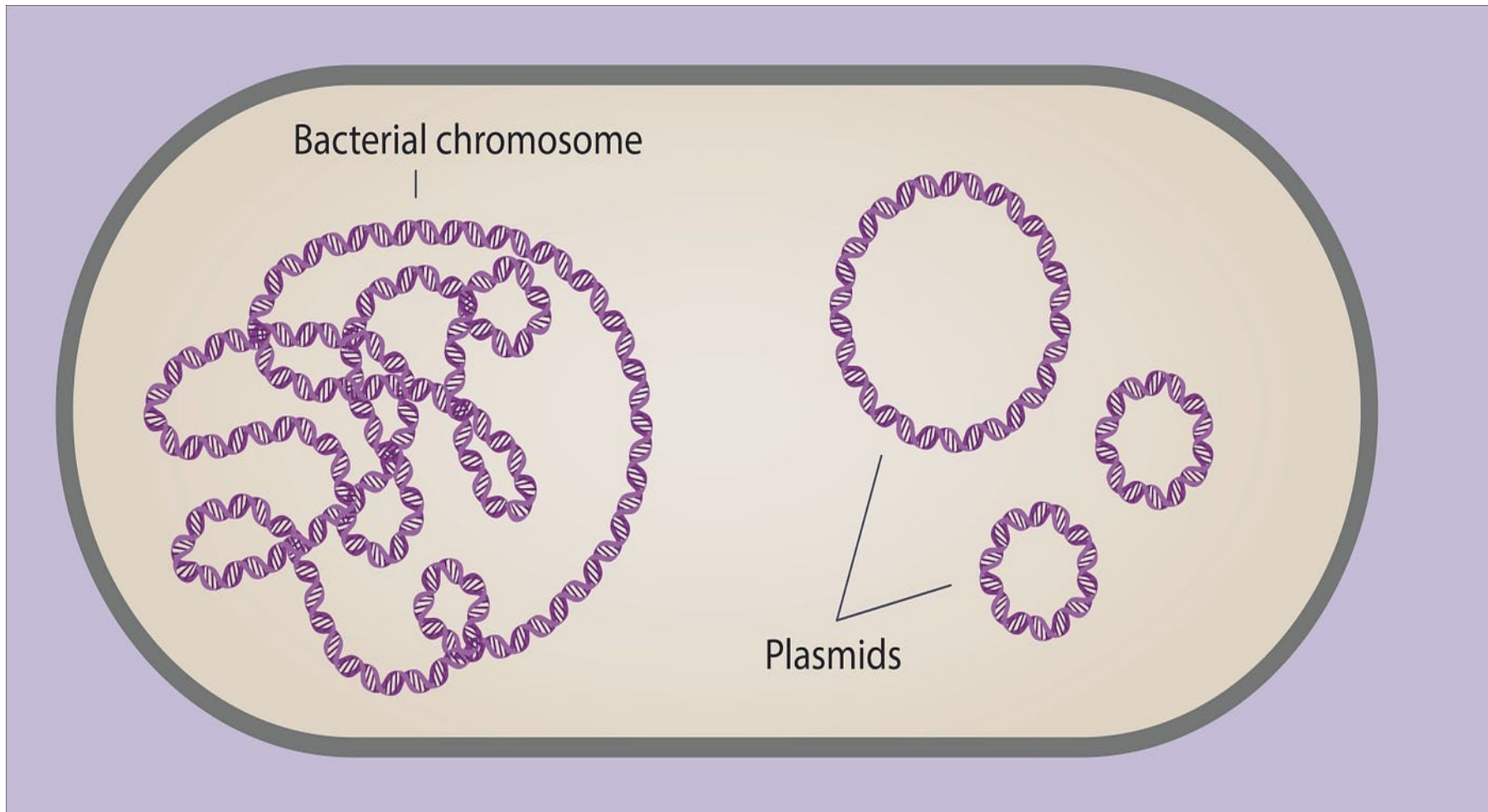
- Typically circular DNA molecule.
- May include one or more plasmids (small, extrachromosomal circular DNA molecules).
- Some bacteria have linear genomes or linear plasmids (e.g., *Borrelia* species).

Size:

- Compact and small compared to eukaryotic genomes.
- Genome sizes range from 0.5 million to 10 million base pairs (Mbp).

Genes:

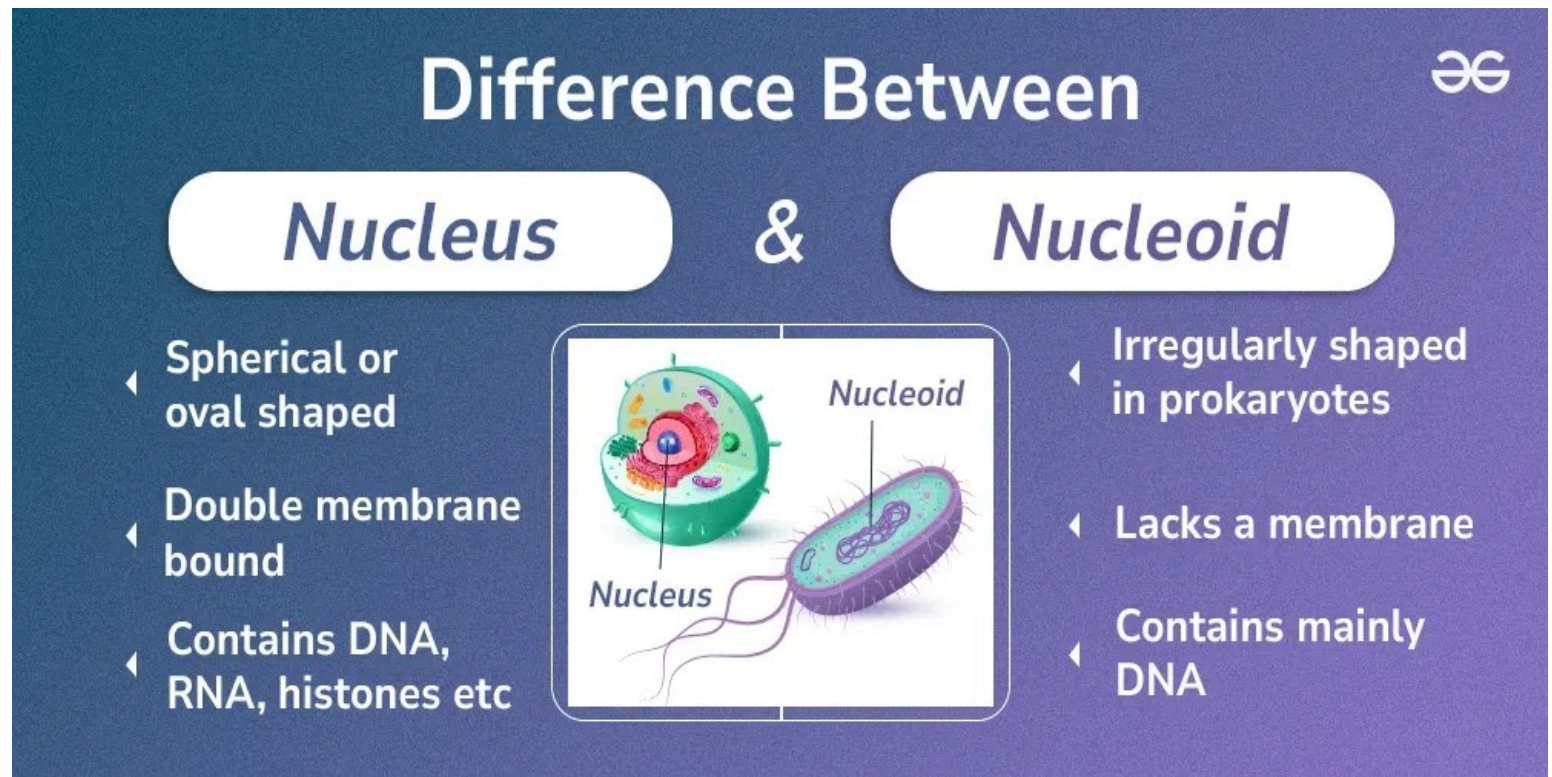
- Encodes around 500 to 5000 genes.
- Genes are arranged in **operons** (groups of genes transcribed together) for coordinated expression.
- Lack of introns in most bacterial genes.



Prokaryotic Genome: Double-Stranded Circular DNA

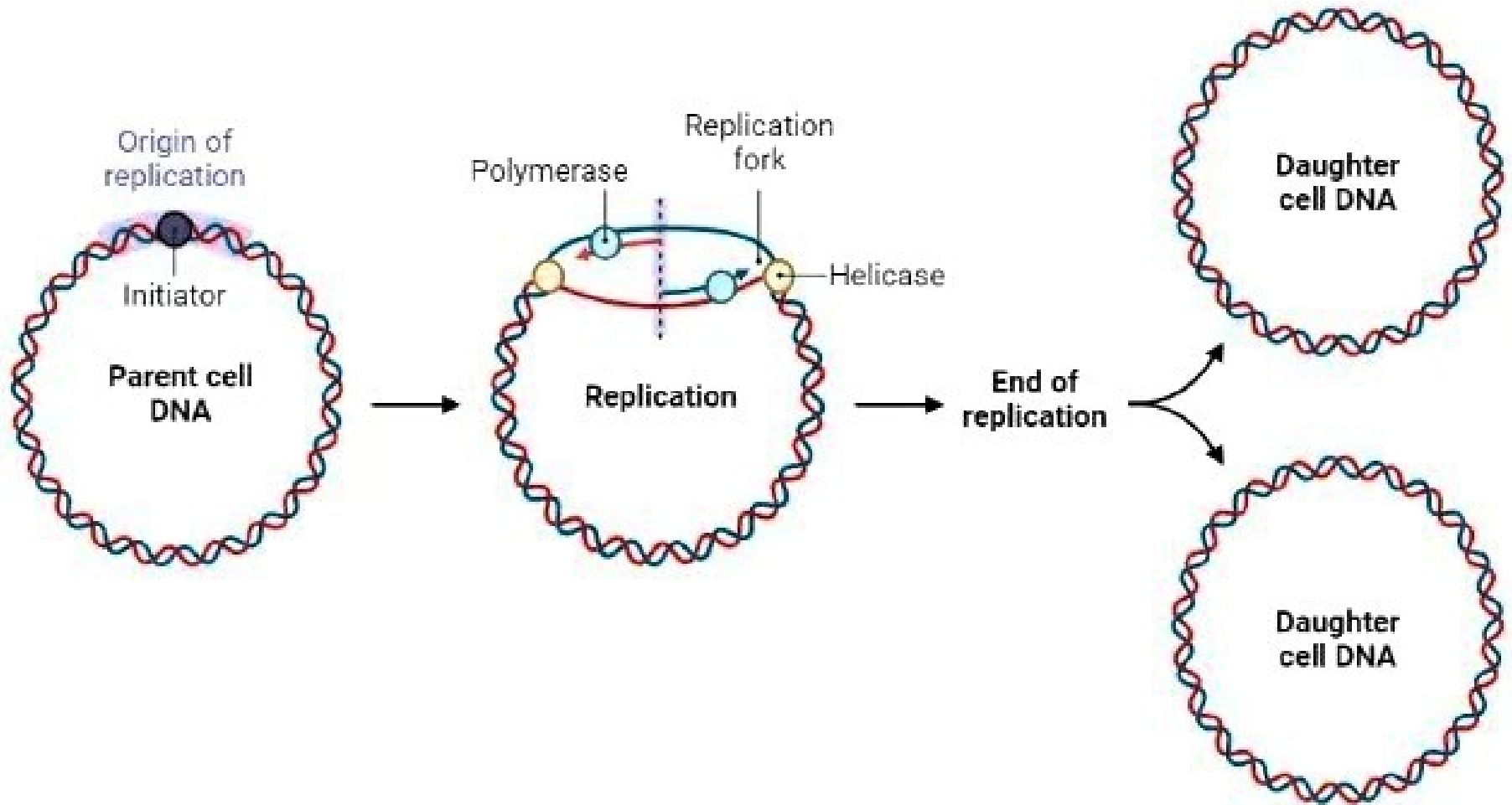
Bacterial Nucleoid

- The nucleoid (meaning nucleus-like) is an irregularly-shaped region within a prokaryote that contains all or most of the genetic material.
- In contrast to the nucleus of a eukaryotic cell, it is not surrounded by a nuclear membrane.

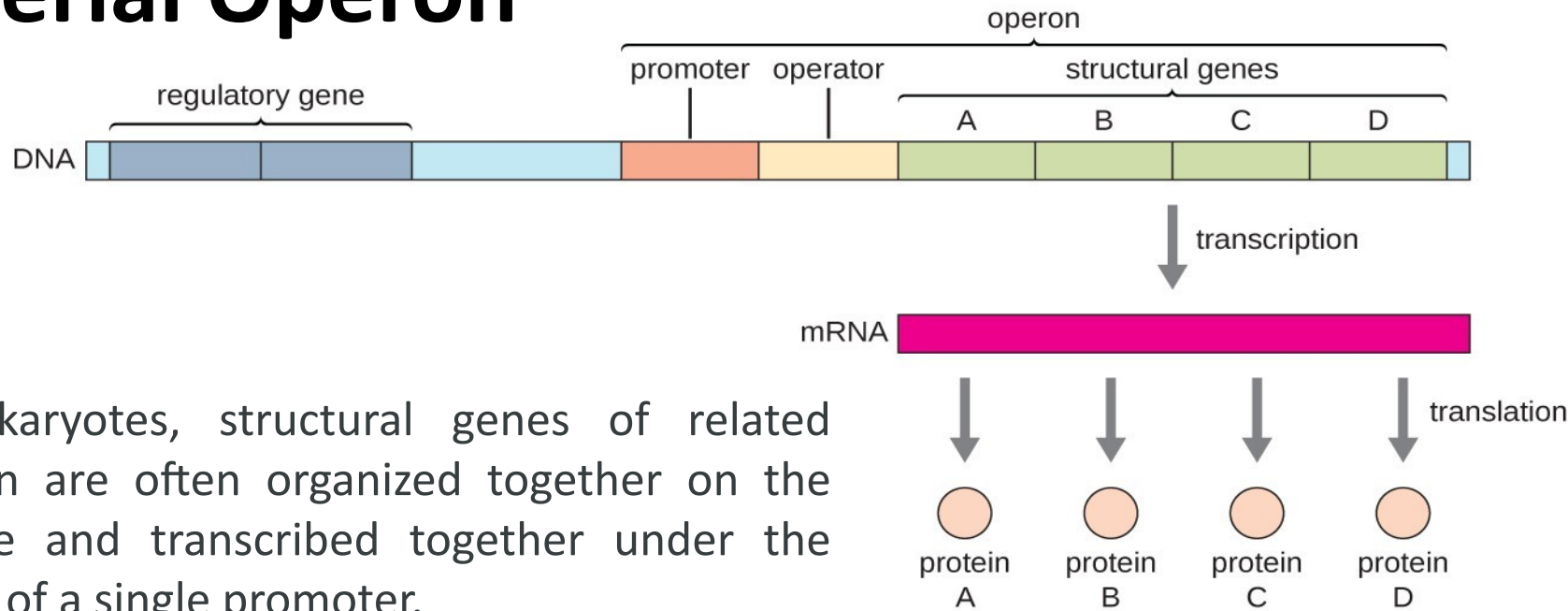


DNA REPLICATION

Prokaryotic DNA replication during cell division



Bacterial Operon



In prokaryotes, structural genes of related function are often organized together on the genome and transcribed together under the control of a single promoter.

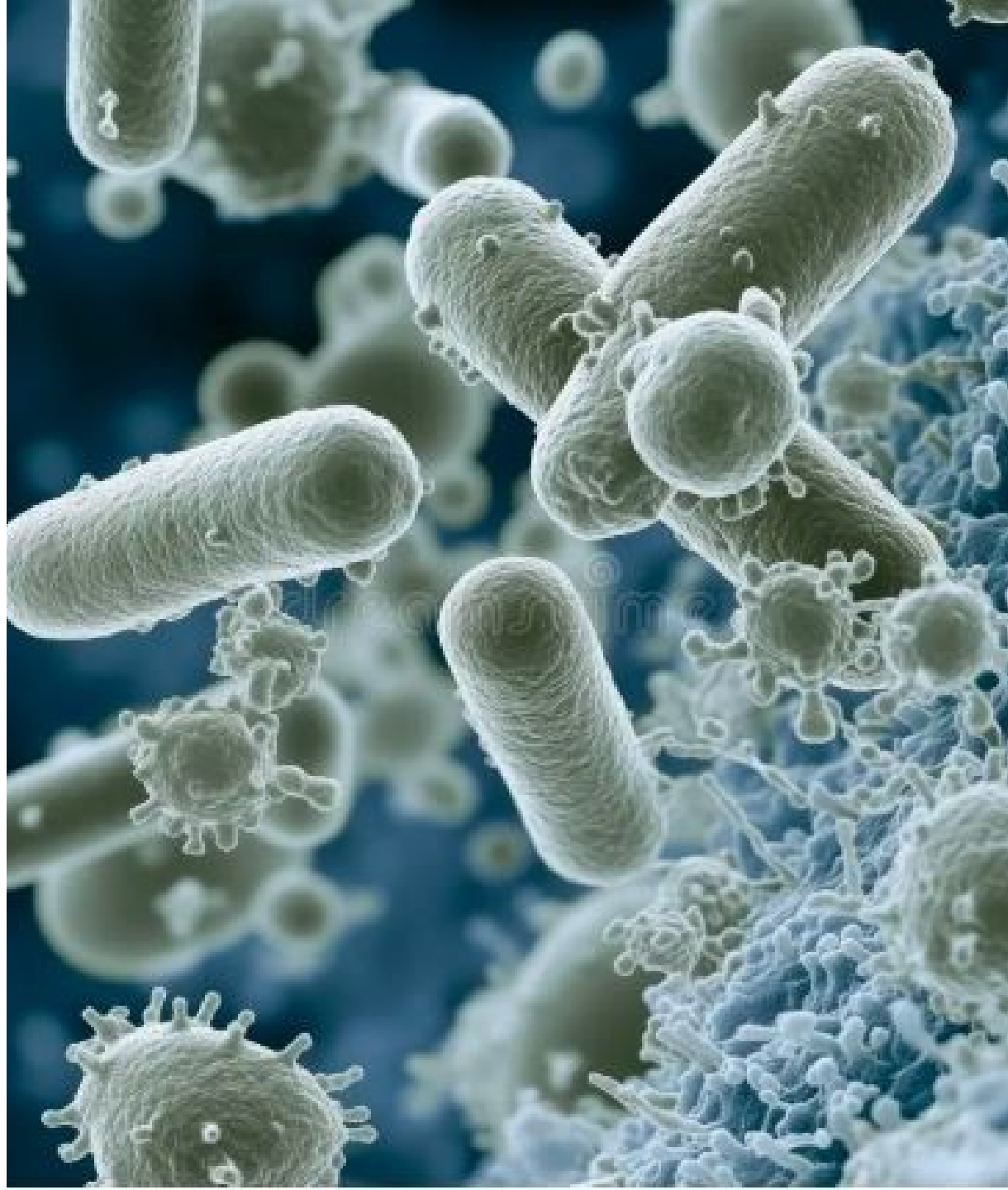
The operon's regulatory region includes both the promoter and the operator.

- If a **repressor** binds to the operator, then the structural genes will not be transcribed.
- Alternatively, **activators** may bind to the regulatory region, enhancing transcription

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Bacterial Genome

Replication:

- DNA replication starts at a single origin of replication (OriC) and proceeds bidirectionally.
- High efficiency and speed compared to eukaryotes.

Organization:

- No membrane-bound nucleus; genome located in the nucleoid region.
- DNA is highly supercoiled and associated with nucleoid-associated proteins (e.g., HU, Fis) for compacting.

Extrachromosomal DNA:

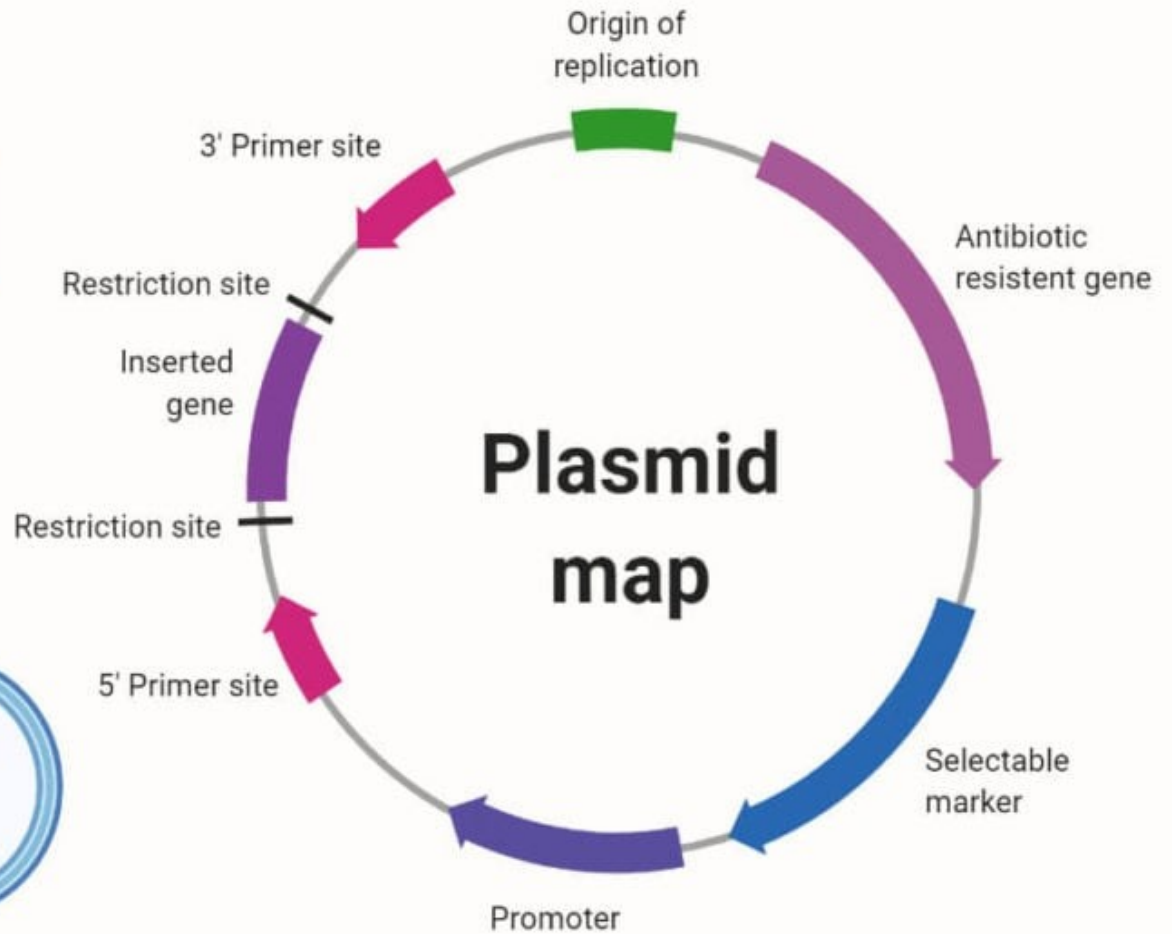
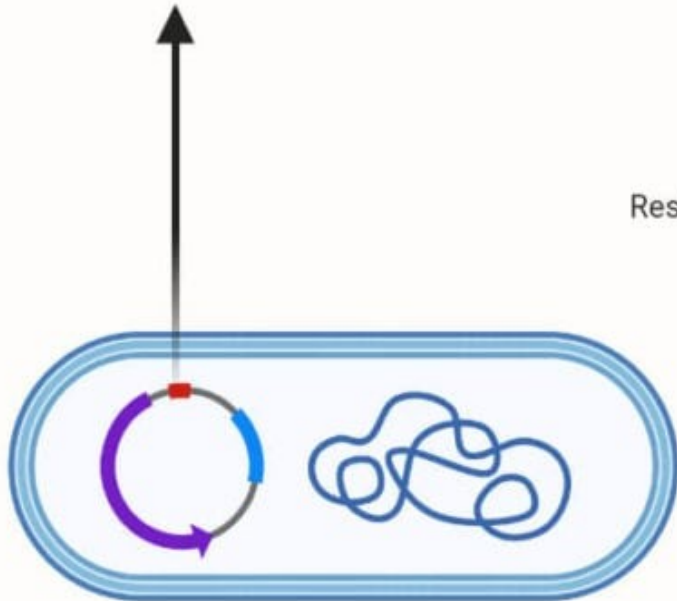
- Plasmids often carry genes for antibiotic resistance, virulence, or metabolic traits.
- Plasmids can be exchanged between bacteria via horizontal gene transfer (e.g., conjugation).

Bacterial Plasmids

Plasmid: A bacterial plasmid is a small, circular, double-stranded DNA molecule ranging from 2000 to 100,000 base pairs that exists independently of the chromosomal DNA in bacteria.

- **Independence from Chromosome:** Plasmids replicate independently of the bacterial chromosome, although they rely on the host's replication machinery.
- **Non-Essential Genes:** Plasmids typically carry genes that are not essential for bacterial survival but provide advantages under specific conditions. Examples include: Antibiotic resistance genes. Virulence factor genes. Genes for metabolizing unusual substances.
- **Transferability:** Many plasmids can transfer between bacteria through processes like conjugation, contributing to the spread of traits like antibiotic resistance.

Plasmid



Types of Plasmids

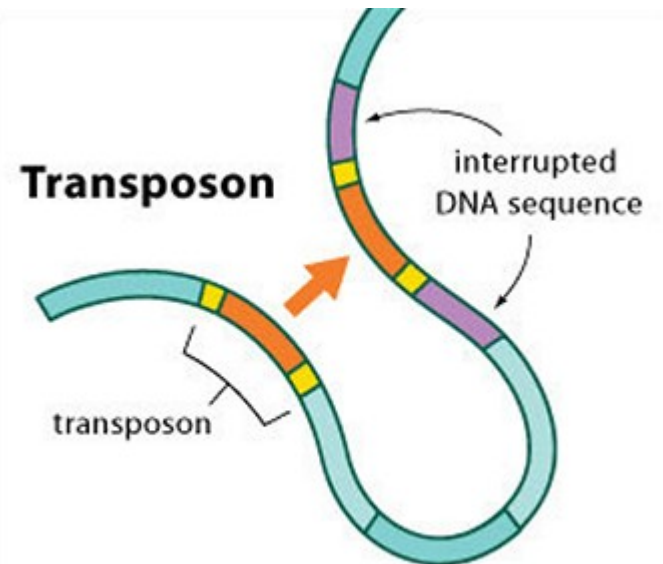
- **Fertility (F) plasmids:** Involved in conjugation (e.g., F plasmid in *E. coli*).
- **Resistance (R) plasmids:** Carry antibiotic resistance genes.
- **Col plasmids:** Produce bacteriocins, which kill other bacteria.
- **Degradative plasmids:** Enable bacteria to metabolize unusual compounds.
- **Virulence plasmids:** Contain genes that enhance pathogenicity.

Plasmids are widely used as vectors in molecular biology and genetic engineering for cloning, gene expression, and producing therapeutic protein

Bacterial Transposons

Bacterial transposons, also known as transposable elements (TEs) or "jumping genes," are DNA sequences that can move from one location to another within a bacterial genome or between DNA molecules such as plasmids and the chromosome.

They play a significant role in bacterial evolution by facilitating genetic variation and the transfer of important traits like antibiotic resistance.



Structure of Transposons

- **Inverted Repeats (IRs):** Short, identical or nearly identical sequences found at both ends of the transposon. They are recognized by the transposase enzyme.
- **Transposase Gene:** Encodes the enzyme responsible for cutting and reinserting the transposon into new DNA sites.
- **Accessory Genes:** Many transposons carry additional genes, such as antibiotic resistance genes or virulence factors.

Types of Transposons

- Insertion Sequences (IS elements):
 - The simplest type of transposon.
 - Contain only the transposase gene and flanking inverted repeats.
- Composite Transposons:
 - Consist of two IS elements flanking a central region containing accessory genes.
 - Example: Tn5, which carries a kanamycin resistance gene.
- Non-Composite Transposons:
 - Lack IS elements and have their own terminal inverted repeats.
 - Example: Tn3, which carries a beta-lactamase gene (conferring ampicillin resistance).
- Conjugative Transposons:
 - Can move between bacteria via conjugation and integrate into the recipient genome.
 - Example: Tn916, which carries a tetracycline resistance gene.

Mechanisms of Transposition

- **Cut-and-Paste Transposition:** The transposon is excised from its original location and inserted into a new site.
- **Replicative Transposition:** The transposon is copied, and the new copy is inserted into a different location, leaving the original in place.

Biological Significance:

- **Genetic Diversity:** Transposons promote genetic rearrangements like deletions, insertions, and inversions.
- **Antibiotic Resistance:** They often carry resistance genes, aiding in the rapid spread of antibiotic resistance between bacteria.
- **Horizontal Gene Transfer:** Transposons can facilitate the transfer of genetic material between different bacteria.

Bacterial Genome

Regulation:

- Gene expression is tightly regulated by mechanisms such as transcriptional repressors, activators, and small RNAs.
- Operons, such as the lac operon, are classic examples of bacterial gene regulation.

Horizontal Gene Transfer:

Bacteria acquire genetic material through:

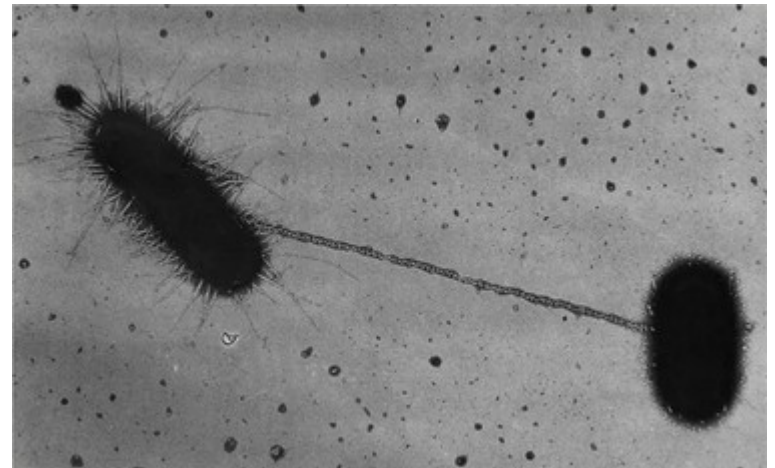
- **Conjugation** (direct transfer via pili).
- **Transformation** (uptake of free DNA from the environment).
- **Transduction** (DNA transfer mediated by bacteriophages).

Mutation and Evolution:

- High mutation rates combined with rapid reproduction drive evolution
- Adaptation to environmental pressures such as antibiotics.

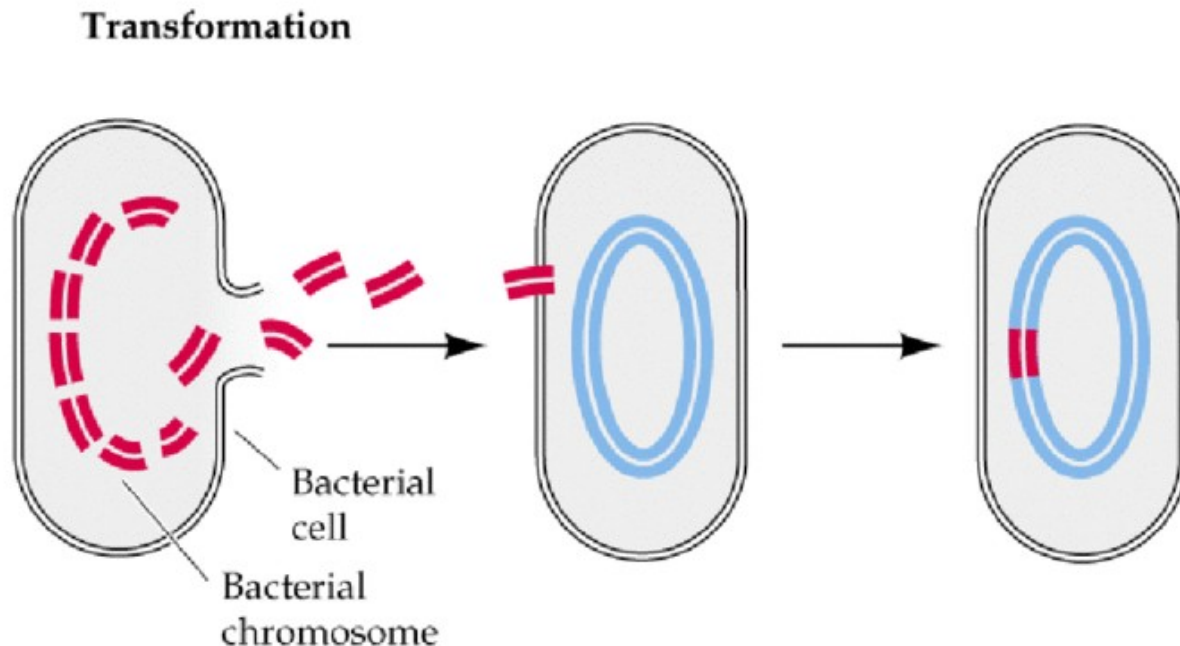
Conjugation

- Conjugation is a gene transfer process in which a recipient bacterium receives DNA from a donor bacterium by cell-to-cell contact through conjugative pili.
- Conjugation is mediated by certain plasmids or transposons
 - Pilus Formation in the donor cells (F+ cells)
 - Physical Contact between Donor (F+ cell) and Recipient (F- cell)
 - Transfer of F-Plasmid
 - Synthesis of Complementary Strand



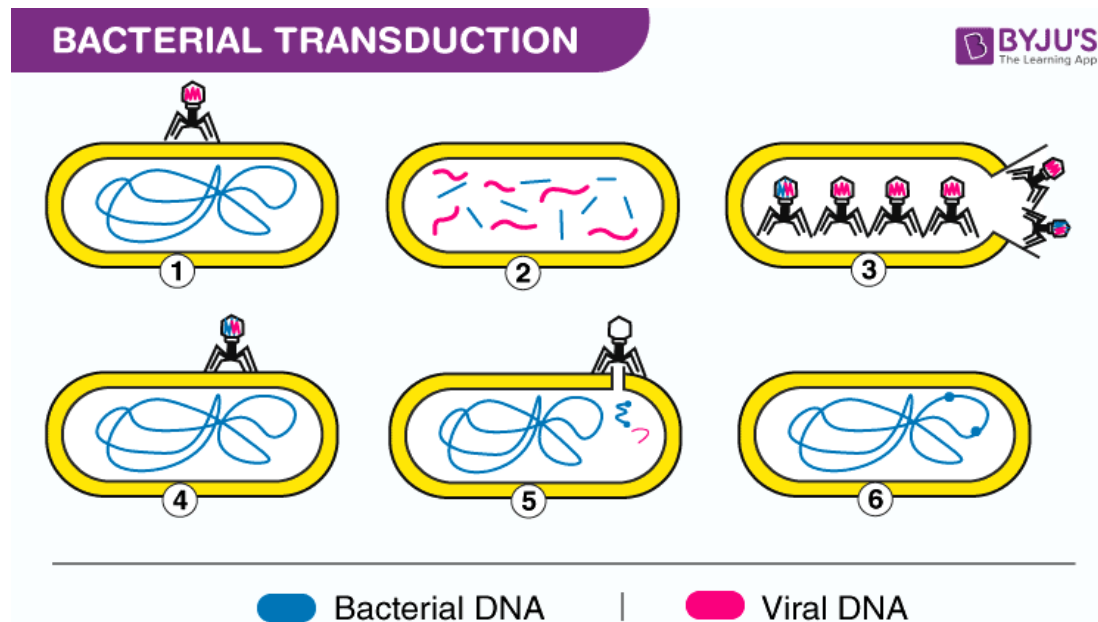
Transformation

- The uptake of DNA molecules through the cell wall from the external surroundings, followed by stable incorporation into the recipient genome, or replication as an independent plasmid.
- Transformation occurs naturally in some species of bacteria, but it can also be done by artificial means in other cells



Transduction

- Transduction is the process by which a virus transfers genetic material from one bacterium to another.
 - **Generalized transduction:** the bacteriophages can pick up any portion of the host's genome.
 - **Specialized transduction:** the bacteriophages pick up only specific portions of the host's DNA



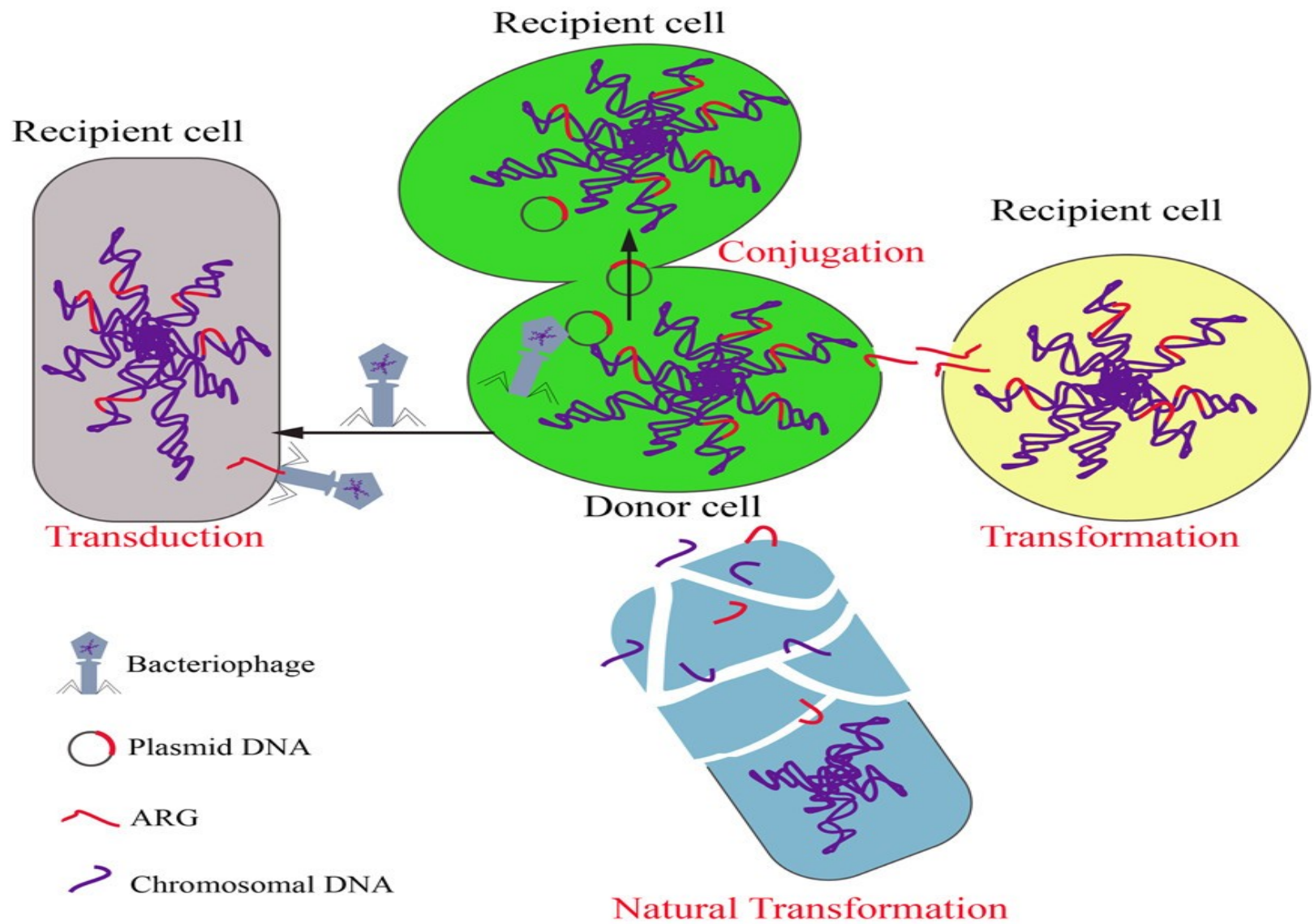
Genetic Basis of Antimicrobial Resistance

Resistance in bacteria can arise through two primary mechanisms:

- **Intrinsic resistance:** Naturally occurring resistance due to inherent structural or functional characteristics
- **Acquired resistance:** Resistance gained through genetic mutations or horizontal gene transfer (HGT).

Genetic Mechanisms of Resistance Spread:

- **Clonal expansion:** A resistant bacterial strain proliferates and becomes dominant in a population.
- **Pandemic clones:** Multidrug-resistant clones like *Klebsiella pneumoniae* ST258 or *E. coli* ST131 spread globally.
- **Reservoirs of resistance:** Environmental bacteria, agricultural settings, and healthcare environments act as reservoirs for resistance genes.



Horizontal transfer of antibiotic resistance genes